Oluwatosin Ayeni

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Anthropic

Dear Hiring Team,

I am applying for the Staff Software Engineer, Inference role at Anthropic. I have spent two years building the kind of inference infrastructure this role is designed to scale — GPU serving stacks, batching pipelines, KV cache optimisation, multi-provider routing, and Kubernetes-based deployment — and I have done it while using Claude as my primary LLM provider.

Three achievements most relevant to the scope of this role:

- **Distributed inference at H100 scale:** I benchmarked vLLM v0.30 vs SGLang v0.5.20 on 2× NVIDIA H100 NVL 94 GB (NVLink 4.0, CUDA 13.4), Qwen3-8B BF16, TP=2. Measured TTFT, TPOT, throughput, and SLO compliance across concurrency 1–128. Peak: 5,965 tok/s. RadixAttention: 1,443× TTFT speedup (39 ms vs 55,565 ms) at 131K-token context. Every result is published live at inference-web-theta.vercel.app/benchmarks/llm.
- **Production inference gateway — batching, caching, routing:** I built *inference-optimizer* — parallel execution (3.87× faster), prompt caching (61.4% cost reduction), model routing (64.3% cheaper), dynamic concurrency (14× throughput) across Claude, GPT-4o, and Gemini. These architecture decisions map directly to the batching, routing, and caching challenges in serving Claude at scale.
- **Kubernetes GPU serving infrastructure:** I built *vllm-k8s-stack* — production LLM serving on Kubernetes with KV-cache-aware routing, HPA autoscaling on GPU node pools, and Prometheus/Grafana observability. Also profiled CUDA KV cache kernels with NVIDIA Nsight, identifying and fixing a memory bandwidth bottleneck that delivered 3.8× throughput improvement.

Anthropic's mission to build safe and beneficial AI is one I take seriously — I use Claude in production and I have written about inference engineering publicly. The Staff role here is a chance to work on the infrastructure that underpins that mission at scale. I am open to relocating to London and available to interview at your convenience.

Yours sincerely,

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